ABSTRACT

Counterfeit currency continues to be a practical problem that affects daily transactions, financial confidence, and economic safety. Even though modern currency notes include several physical security features, fake notes still enter circulation due to improved printing techniques and the difficulty faced by common users in identifying them. In many cases, verification depends on manual checking or centralized digital systems, both of which have limitations related to accuracy, transparency, and data control.

This project presents a web-based prototype that focuses on verifying currency notes using a combination of image processing, serial number extraction, and blockchain-based record storage. The system allows a user to upload an image of a currency note, after which Optical Character Recognition is applied to extract the serial number printed on the note. Basic image enhancement steps are used to improve readability before verification is performed.

Instead of relying on a single database, the extracted serial number is checked against verification records stored on a blockchain ledger. Each verification event is recorded using cryptographic hashing, creating a permanent entry that cannot be modified later. If the serial number already exists, the system flags the currency as duplicate or suspicious. Otherwise, the note is treated as genuine and a new record is added.

An administrative interface is also included to observe verification activity and review stored records. By avoiding centralized control and maintaining a permanent verification history, the system improves trust and accountability in currency verification. The prototype demonstrates that blockchain can function as a reliable backend for recording and validating currency-related data, while leaving scope for future improvement and real-world adaptation.

CHAPTER 1

INTRODUCTION

1.1 Definition

Currency verification, in practical terms, involves determining whether a physical currency note is authentic or has been previously duplicated through unauthorized means. In traditional systems, this judgment is often based on visible security features and manual inspection, which can vary in accuracy depending on the observer and conditions.

In this project, currency verification is approached as a digitally assisted validation process. An image of the currency note is used as the primary input, from which the serial number is extracted using Optical Character Recognition techniques. This extracted identifier acts as the key reference for verification rather than relying solely on physical observation.

1.2 Objectives

The primary objective of this project is to design and implement a secure, transparent, and automated currency verification system using blockchain framework. The detailed objectives are as follows:

- To develop an automated currency verification process that reduces dependence on manual inspection and minimizes errors caused by human judgment.
- To extract serial numbers from currency note images accurately by applying Optical Character Recognition techniques, even when images vary in lighting, orientation, or clarity.
- To maintain currency verification records on a blockchain ledger so that once a verification entry is created, it remains permanent and traceable.
- To prevent unauthorized modification or duplication of verification data by applying cryptographic hashing mechanisms.
- To detect whether a currency note has been verified previously and classify it appropriately as genuine, duplicate, or suspicious.
- To create a reliable audit trail of all verification activities, enabling historical tracking of currency verification events.

1.3 Problem Statement

The circulation of counterfeit currency continues to pose a serious challenge to economic stability and public confidence. Although modern currency notes include several physical security features, counterfeit notes still manage to enter circulation due to advances in printing technology and the difficulty faced by ordinary users in identifying fake currency accurately.

Existing verification largely approaches depend on manual inspection or centralized digital systems. Manual methods are subjective, time-consuming, and prone to inconsistency, while centralized databases introduce risks such as data manipulation, lack of transparency, and single-point failure. Once compromised, centralized systems limited offer assurance regarding the integrity of stored verification records.

1.4 Existing System

The existing currency verification systems can be broadly categorized into manual methods and centralized digital systems:

Manual Inspection Methods

- Rely on physical features such as watermarks, security threads, and colourshifting ink.
- Require trained personnel and are prone to human error.

Centralized Digital Verification Systems

- Store verification data in centralized servers or databases.
- Vulnerable to hacking, data tampering, and insider manipulation.

Limitations of Existing Systems

- No cannot be modified record of verification history.
- Difficulty in detecting duplicate circulation of the same currency note.

1.5 Proposed System

The developed system introduces a secure and decentralized approach for currency verification using blockchain framework. It is designed to overcome the limitations of manual inspection and centralized verification systems by providing an automated, resistant to alteration, and transparent authentication mechanism. The system allows users to verify currency authenticity using a simple web-based interface without requiring specialized hardware or expert knowledge.

In the proposed approach, users upload an image of the currency note through the application interface. The system first validates the image to ensure it meets required quality and format standards. This step prevents invalid or unreadable inputs from entering the verification workflow and provides reliable processing in subsequent stages.

1.6 Scope of The Project

The scope of this project is limited to the design and development of a blockchain-supported system for verifying Indian currency notes using serial number information. The system focuses on extracting serial numbers from uploaded currency images through Optical Character Recognition and validating them against blockchain-stored verification records. Based on the verification outcome, currency notes are categorized as genuine, duplicate, or suspicious. An admin dashboard is included to monitor verification logs and review system activity.

CHAPTER 2

LITERATURE SURVEY

[1] S. Kumar and R. Verma (2020) – Image-Based Currency Authentication Using Feature Extraction

Currency authentication is a critical concern in financial systems due to the increasing circulation of counterfeit banknotes. This study focuses on using image processing techniques to verify physical currency by extracting visual features such as watermarks, security threads, and texture patterns. The proposed approach applies preprocessing methods including grayscale conversion, edge detection, and feature matching to distinguish genuine notes from counterfeit ones. Experimental results indicate moderate accuracy under controlled conditions. However, the system is sensitive to lighting variations, note quality, and camera resolution.

[2] A. Sharma and P. Mehta (2021) – OCR-Based Serial Number Detection for Banknote Verification

This work introduces an Optical Character Recognition (OCR) based system for extracting serial numbers from currency notes. The extracted serial numbers are compared against a predefined dataset to identify invalid or duplicate notes. The approach improves automation and reduces human effort in manual verification. While OCR enhances speed and usability, the system heavily depends on image clarity and font consistency. Errors occur when notes are worn, folded, or partially damaged. More importantly, the stores system serial numbers in a centralized database, making it vulnerable to data tampering, unauthorized modification, and single-point failure.

[3] R. Patel et al. (2021) – Centralized Digital Currency Tracking System Using Databases

This paper proposes a centralized digital tracking system where currency serial numbers are registered and verified through a central authority. Each note’s serial number is stored in asecure server, and verification is performed by querying the database. The system improves traceability and enables rapid verification in controlled environments such as banks. However, centralized storage introduces trust issues, as the authority maintaining the database has full control over data modification.

[4] M. Singh and K. Kaur (2022) – Blockchain-Based Anti-Counterfeiting Framework for Financial Assets

This study explores the use of blockchain framework to address trust and security issues in asset verification systems. By storing verification records on a decentralized ledger, the framework provides immutability and clear verification trail. Smart contracts are used to automate validation logic and prevent unauthorized updates. The system successfully eliminates centralized control and enhances trust among participants. However, the work is largely conceptual and focuses on digital assets rather than physical currency.

[5] L. Ahmed and Y. Hassan (2023) – Smart Contract–Driven Verification System for Secure Transactions

This paper presents a smart contract–based verification mechanism implemented on the Ethereum blockchain. The system validates asset identifiers and records verification events immutably. Automated contract execution removes human intervention and enforces consistent verification rules. While the system demonstrates strong security properties, it assumes that input data is already trusted. The study does not consider how physical identifiers such as currency serial numbers are captured or validated before submitted being to the blockchain. Additionally, performance constraints and scalability in high-frequency verification scenarios are identified as limitations.

[6] Proposed System – Blockchain-Based Currency Verification with OCR and Admin Monitoring

The developed system integrates OCR-based serial number extraction with blockchain-backed verification to authenticate physical currency notes securely. Serial numbers extracted from currency images are verified against cannot be modified records stored in smart contracts, ensuring authenticity and preventing duplication. A hybrid verification approach combines fast local checks with authoritative blockchain validation to balance performance and security. The system also includes an admin dashboard for monitoring verification logs, suspicious activity detection, and audit reporting.

Ref	Paper Title	Objective	Methodology	Research Gap
[1]	Image-Based Currency Authentication Using Feature Extraction (2020)	To identify counterfeit currency using visual features	Image preprocessing, feature extraction, and visual pattern matching	Sensitive to lighting and image quality; no secure record storage
[2]	OCR-Based Serial Number Detection for Banknote Verification (2021)	To automate currency verification using serial numbers	OCR extraction and comparison with centralized database	OCR errors on damaged notes; centralized data vulnerable to tampering
[3]	Centralized Digital Currency Tracking System (2021)	To track and verify currency using a central authority	Serial number registration and database querying	Lack of clear verification trail; single-point failure; limited scalability
[4]	Blockchain-Based Anti-Counterfeiting Framework (2022)	To improve trust in asset verification	Blockchain ledger and smart contracts	Focused on digital assets; ignores physical currency challenges
[5]	Smart Contract–Driven Verification System (2023)	To ensure resistant to alteration verification using blockchain	Ethereum smart contracts for identifier validation	Assumes trusted input; performance and scalability issues
[6]	Proposed System: Blockchain-Based Currency Verification	To securely verify physical currency notes	OCR-based serial extraction + blockchain verification + admin monitoring	Prototype-level implementation; depends on OCR accuracy

CHAPTER 3

REQUIRMENT ANALYSIS

Requirement analysis is actually the most crucial stage in the overall developing activity of any software; it primarily defines what the system has to perform and how it should perform under various conditions. This Analysis phase currently takes the Currency Verification Using Blockchain framework project under intense scrutiny to get the functional needs, data requirements, and modes of operation associated with the proposed solution. This phase tells that the system does so in such a way that the process of authenticating currency authenticity is carried out very efficiently within the following parameters-insecurity, verification track, and reliability at the end.

It consists of image processing, OCR, blockchain framework, and web based interfaces used for developing a system. So it has to be kept in mind the thorough understanding of how all this will function as a whole, along with which resources, for a smooth environment.

3.1 Problem Analysis

3.1.1 Background of the Problem

Counterfeit currency is still a serious menace to national economics, banking systems, and public confidence. Advanced machines may be for used the verification of banks and forensic laboratories, but they cannot be accessed by common people.

Conventional verification procedures depend absolutely on visual inspection, ultraviolet light checks, or centralized databases maintained by financial institutions. All these mechanisms are subjective, expensive or fallible.

3.1.2 Existing Challenges

The major challenges identified in existing currency verification systems are listed below:

- Difficulty for common users to verify currency authenticity without specialized devices
- Dependence on centralized databases that can be altered or attacked
- Lack of permanent and auditable verification records

3.1.3 Need for the developed system

The developed system addresses the above challenges by integrating OCR-based serial number extraction with blockchain-based verification storage. By doing so, the system provides:

- Automated extraction of currency serial numbers from images
- Permanent storage of verification records on blockchain
- Detection of duplicate or suspicious currency usage
- Transparent and resistant to alteration audit logs
- Easy access through a web interface for users and administrators

3.2 Data Understanding

Data Understanding Play a Crucial Role in Designing of a Verification System. This Project to be understanding Data that Consist of Image-based Inputs and Records stored on Blockchain. Understanding This Data Nature, Structure, and Its Challenges Are Such That an Accurate Processing and Verification Is Maintained.

3.2.1 Nature of the Data

- Currency images are generally considered unstructured visual data that vary in illumination, angle, background, and resolution.
- Serial numbers must be accurately extracted from the well-structured alphanumeric data so they can be verified.
- As the name suggests, blockchain data cannot be changed, has a timestamp, and is distributed at all places, thus maximizing integrity.
- Verification logs are semi-structured records used for monitoring and auditing.

3.2.3 Data Challenges

The system faces several data-related challenges:

- OCR accuracy may be affected by poor image quality or damaged notes
- Variations in font, spacing, and colour of serial numbers
- Duplicate serial numbers may appear in multiple verification attempts
- Blockchain transaction latency and gas cost considerations

3.2.4 Data Preparation Requirements

To ensure reliable system performance, the following data preparation steps are required:

- Image preprocessing such as resizing, grayscale conversion, and noise reduction
- OCR text cleaning to remove unwanted characters
- Hashing serial numbers before storing them on the blockchain

3.3 Input Analysis

The system accepts the following inputs:

- Currency notes image (PNG/JPG format)
- User authentication credentials
- Verification request initiated by the user
- Optional metadata such as location and device information

Each input is validated to prevent malicious or incorrect data from entering the system.

3.4 Output Analysis

The outputs generated by the system include:

- **Verification Result**

  Displays whether the currency is Genuine, Duplicate, or Suspicious.

- **Proof**

  Shows transaction hash, block number, and timestamp.

- **Verification Details**

  Displays extracted serial number, currency value, verification time, and location.

- **Admin Dashboard** provides a detailed list of all verification activities with receipt download options.

- **Confirmation**

  Green or red status indicators for easy understanding by users.

CHAPTER 4

SYSTEM REQUIREMENT SPECIFICATION

System Requirement Specification (SRS) defines the technical and operational requirements needed to develop and deploy the proposed Currency Verification Using Blockchain framework system.

4.1 Hardware Requirements

The hardware requirements specify the minimum and recommended resources required to run the system smoothly during development and deployment.

Processor

- Intel Core i5 or equivalent AMD Ryzen processor
- Recommended: Intel Core i7 / AMD Ryzen 7

A multi-core processor is required to handle OCR processing, blockchain interactions, and concurrent user requests efficiently.

Memory (RAM)

- Minimum: 8 GB
- Recommended: 16 GB

Adequate memory is required for image preprocessing, OCR execution, and maintaining backend services without performance degradation.

1. Storage

- Minimum: 20 GB free disk space

Storage is required for:

- Uploaded currency images
- OCR output files
- Blockchain logs and receipts
- Application source code and dependencies

2. Network Requirements

- Stable internet connection (minimum 5 Mbps)

Internet access is required for:

- Blockchain network communication
- Smart contract interaction
- Admin dashboard access

3. Peripheral Devices

- Digital camera or smartphone (for capturing currency images)
- Standard display monitor

These devices help users capture and view currency verification results clearly.

4.2 Software Requirements

Software requirements define the operating environment and tools needed to build and execute the system.

1. Operating System

- Windows 10 / 11

The system is platform-independent and can run on any OS supporting Python and web technologies.

2. Programming Languages

- Python – backend logic, OCR, blockchain interaction
- JavaScript – frontend interaction
- HTML and CSS – user interface design

3. Development Frameworks

Backend Framework

- Flask or Django (Python-based web framework)

Frontend Framework

- HTML, CSS, JavaScript

These frameworks enable rapid development and easy integration with blockchain services.

4. OCR and Image Processing Tools

- Tesseract OCR – serial number extraction
- OpenCV – image preprocessing (grayscale, thresholding, noise removal)
- Pillow (PIL) – image handling

5. Blockchain and Smart Contract Tools

- Ethereum Blockchain (Test net / Local Ganache)
- Web3.py – Python interface to Ethereum
- Solidity – smart contract development
- Ganache – local blockchain testing

6. Database and Storage

- SQLite / PostgreSQL – store user data and logs
- File System / Cloud Storage – store images and reports

4.3 Functional Requirements

Functional requirements describe what the system is expected to do.

1. User Authentication

- The system shall allow users to log in securely using valid credentials.
- Unauthorized users shall be denied access.
- Sessions shall be maintained securely until logout or timeout.

2. Currency Image Upload

- The system shall allow users to upload currency images in JPG or PNG format.
- Uploaded files shall be validated for size and format.

- Invalid or unsupported files shall be rejected.

3. OCR-Based Serial Number Extraction

- The system shall extract the serial number from the uploaded currency image using OCR.
- Extracted text shall be cleaned and validated.
- If serial extraction fails, the system shall notify the user.

4. Blockchain Verification

- The system shall check the extracted serial number against blockchain records.
- Each verification request shall generate a blockchain transaction.
- The system shall determine whether the currency is:
- Genuine
- Duplicate
- Suspicious

5. Blockchain Record Storage

- The system shall store verification records on the blockchain.
- Stored data shall include:
- Hashed serial number
- Timestamp
- Currency value

6. Verification Details Display

- The system shall display verification results clearly to the user.
- The verification panel shall show:
- Extracted serial number
- Verification status
- Blockchain proof

        - Date and time

7. Admin Dashboard

- The system shall provide an admin dashboard.
- Admin shall be able to:
    - View all verification logs
    - Track duplicate or suspicious currency
    - Download blockchain receipts
    - Monitor system activity

8. Verification Log Management

- The system shall maintain a permanent log of all verification activities.
- Logs shall be cannot be modified once recorded on blockchain.

4.4 Non-Functional Requirements

Non-functional requirements define how the system performs rather than what it does.

1. Performance

- The system shall process verification requests within acceptable time limits.
- OCR and blockchain verification shall be optimized for faster response.

2. Security

- The system shall protect user data and verification records.
- Blockchain shall ensure resistant to alteration storage.
- Sensitive data shall not be stored in plain text.

3. Reliability

- The system shall function correctly even under repeated verification attempts.
- Blockchain data shall remain consistent across all nodes.

4. Scalability

- The system shall support increased number of users and verification requests.
- Architecture shall allow future cloud deployment.

5. Usability

- The interface shall be simple and easy to understand.
- Verification results shall be displayed using clear labels and indicators.

6. Maintainability

- The system shall be modular and easy to update.
- Smart contracts shall support future upgrades with minimal changes.

Summary of System Requirements

This chapter defined the complete system requirements for implementing the Currency Verification Using Blockchain framework project. By clearly identifying hardware, software, functional, and non-functional requirements, this specification provides that the system is secure, efficient, and scalable. These requirements serve as a strong foundation for system design, implementation, and testing in subsequent chapters.

CHAPTER 5

SYSTEM DESIGN

Introduction to System Design

This chapter presents the detailed methodology adopted for the Currency Verification Using Blockchain project. The methodology is framed to reflect a research-paper–based prototype, focusing on conceptual design, logical workflow, and feasibility validation rather than production-level deployment. The proposed approach integrates conventional verification mechanisms with blockchain-based validation to address trust, integrity, and auditability concerns inherent in centralized systems.

5.1 System Architecture

The developed system architecture follows a hybrid three-tier model consisting of the User Interaction Layer**,** Application Processing Layer**,** and Blockchain Layer**.** This layered design provides separation of concerns, modularity, and clarity in data flow. Each layer performs a specific function within the verification workflow.

The User Interaction Layer provides interfaces for administrators and users to interact with the system. The Application Processing Layer handles verification logic, preprocessing, and coordination between local data and blockchain queries

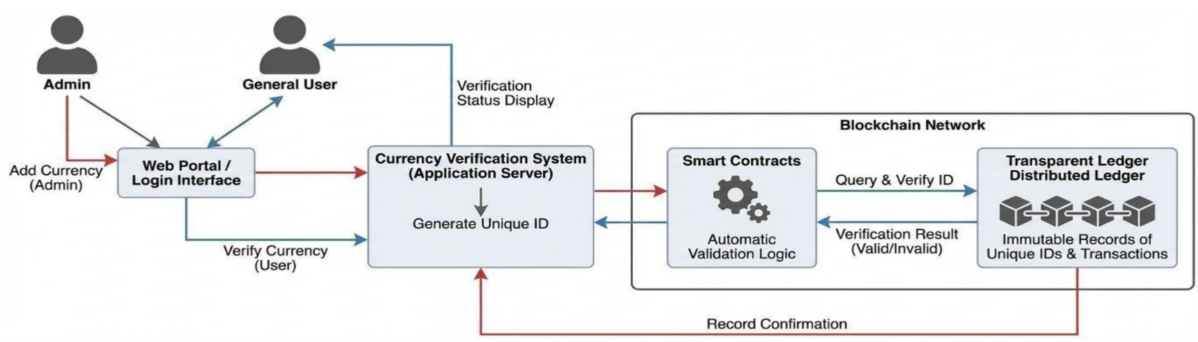

Fig 5.1 Architecture Diagram

1. User / Admin

This component represents the external actors of the system.

- User: Uploads the currency image for verification.
- Admin: Manages system-related operations such as registering valid currency information or maintaining verification records (conceptual role).

They do not interact directly with the blockchain. All interactions are routed through the application system.

2. Web Interface

The Web Interface acts as the entry point to the system.

- Accepts currency images from users
- Displays verification results
- Provides controlled access to system functionality

This layer isolates users from internal processing complexity and blockchain operations.

3. Application Server (OCR + Hotlist + Logic) This

is the core processing layer of the system.

It performs:

- OCR processing to extract the serial number from the currency image
- Preprocessing and validation of extracted data
- Hotlist checking for known counterfeit serials
- Verification logic to decide whether blockchain validation is required

This component provides that only necessary verification requests are forwarded to the blockchain, improving efficiency.

4. Blockchain Network (Smart Contract Layer)

The Blockchain Network provides the trust and immutability layer**.**

- Smart contracts store and verify currency identifiers
- Verification records are cannot be modified and resistant to alteration
- The system queries the blockchain only when required

This layer eliminates reliance on centralized trust and provides transparent verification.

5.2 Data Flow Diagram (DFD – Level 0 / Context Level)

Context Diagram Description:

System (Process 0): Currency Verification Using Blockchain System

The Currency Verification Using Blockchain System represents the primary process responsible for authenticating physical currency using a blockchain-based verification approach. The system accepts currency images submitted by users and performs preliminary verification operations such as serial number extraction, formatting, and validation. Based on the processed input, the system interacts with the blockchain network to verify the authenticity of the currency using cannot be modified ledger records. After completing the verification workflow, the system returns the verification status to the user through the system interface.

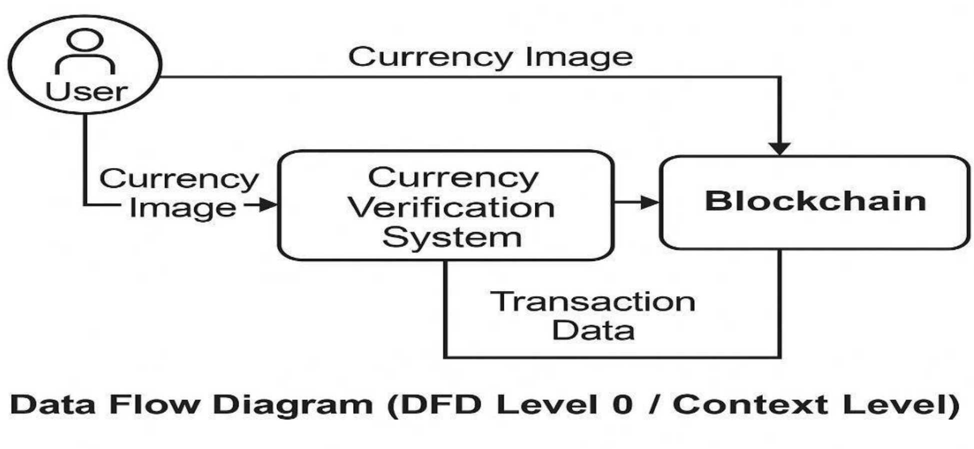

Fig 5.2.1 Data Flow Diagram – Level 0 (Context Level)

External Entity User

The User acts as an external entity who initiates the currency verification workflow. The user uploads an image of the currency note to the system and submits a verification request. After processing, the user receives the verification outcome from the system, indicating whether the currency is genuine or identified as counterfeit.

External Entity Administrator

The Administrator is an external entity responsible for managing currency-related information at a conceptual level. This includes activities such as registering currency identifiers, maintaining verification records, and overseeing system operations.

High-Level Data Flows (Context Level)

- **User → System:**
  Currency Image and Verification Request submitted for authentication.
- **System → User:**
  Currency Verification Result indicating authenticity status.
- **Administrator → System:**
  Currency Registration or Management Data for system maintenance.
- **System → Blockchain Network:**
  Verification Query and Transaction Data for validating currency identifiers.
- **Blockchain Network → System:**
  Verification Response (Valid / Invalid) retrieved from cannot be modified ledger records

5.3 Data Flow Diagram (DFD – Level 1 Detailed)

The Level-1 Data Flow Diagram expands the proposed Currency Verification Using Blockchain System into its internal functional modules. This level provides a detailed view of how data is processed and transformed within the system.

1. Input Module Receiving

Receives Currency Images Uploads by the Users from system UI.

2. Serial Number Extraction Module

Utilizes Optical Character Recognition Techniques to Extract Serial Number from Currency of the Upload Image.

3. Hotlist Verification Module

Compare with the predefined hotlist of identifiers already known to forged or blacklisted against the extracted serial number.

4. Blockchain Verification Module

Starts the validation to the block chain network in case the serial number is not present in the hotlist.

5. Monitoring and Logging Module

Keeps configuring the workflow of verification steps related to the system and the processing status.

6. Output Module

This Module Displaying the Latest Verification Result of a Currency to the User that Defines His or Her Notes as Being Genuine or Counterfeit.

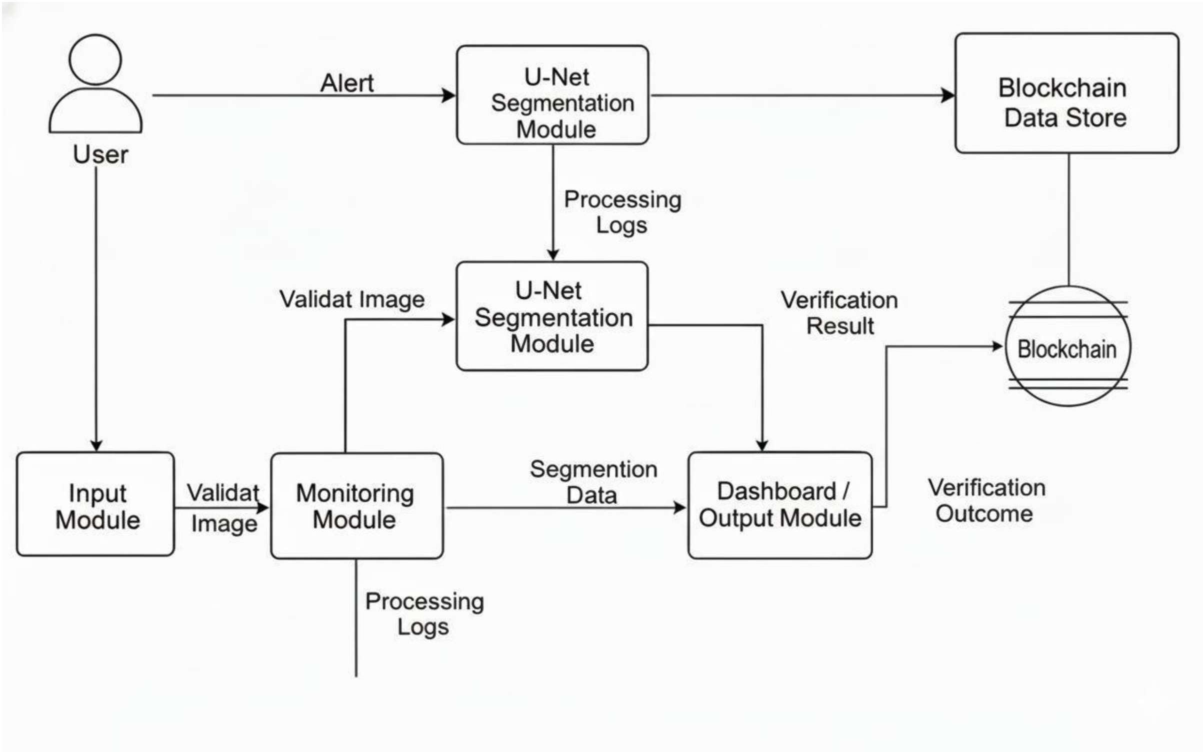

Fig 5.3.1 Data Flow Diagram – Level 1 (DFD – Level 1 Detailed)

Explanation

The Level 1 Data Flow Diagram provides a detailed internal view of the proposed Currency Verification Using Blockchain System by decomposing the overall verification workflow into multiple functional modules.

5.4 Activity Diagram

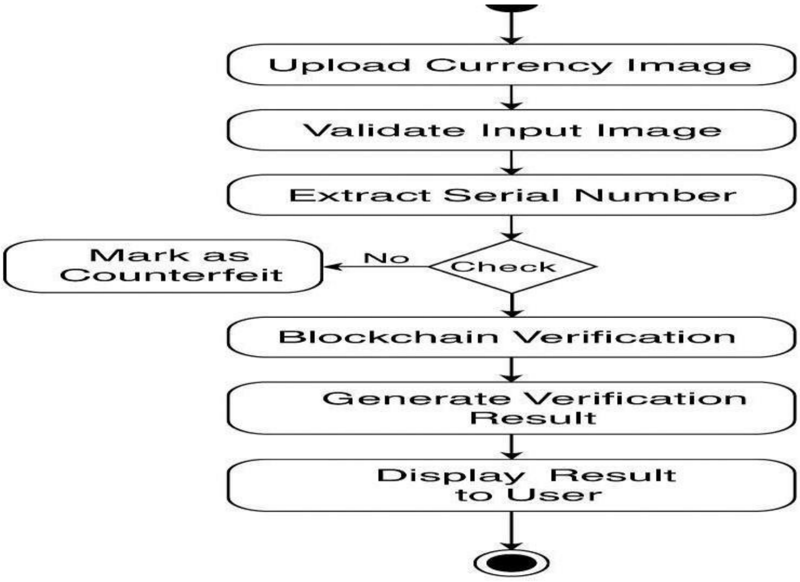

Fig 5.4.1 Activity Diagram

Flow:

1. The login/authentication of a user, in this case, will enable the authenticated user to get into the system using valid user credentials.

2. Currency Picture Upload: The user uploads currency note images on their online services in JPEG or PNG format.

3. Input Validation/Integrity Check: Currency images get into folders for checks of corruption, type of format, and unreadability of content.

4. Serial Number Extraction: The system will use Optical Character Recognition techniques to extract the currency serial number from the image.

5. Decision: Hotlist Check: Retrieved serial number will checked against local hotlist of counterfeit currencies.

6. If it is not blacklisted, the system invokes verification through blockchain.

7. The outcome of verification will be a level of success of whether the currency is truly genuine or counterfeit.

8. The final status would be determined upon aggregating the results of verification from local and blockchain sources.

9. the audit trail would capture the verification, timestamp, serial number, result of verification, and transaction reference.

10. The final display of result to the User would be through interface which declares the currency is Genuine/False.

Explanation

The Activity Diagram illustrates the complete operational workflow of the proposed Currency Verification Using Blockchain System, beginning with user authentication and ending with the display of the final verification result. It highlights how the system processes currency images through a sequence of validation, serial extraction, blacklist checking, and blockchain-based authentication.

5.5 Use Case Diagram

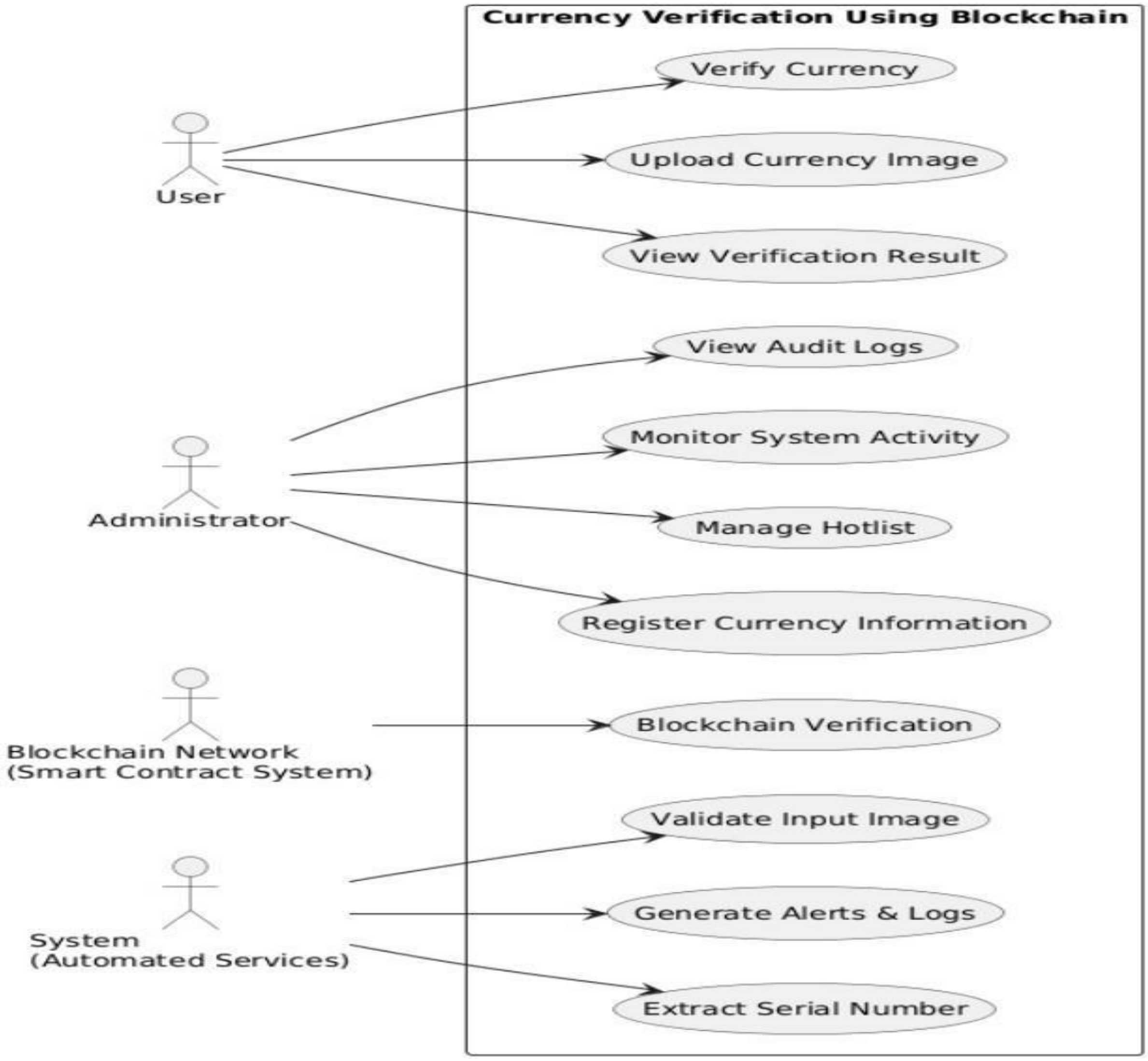

Fig 5.5.1 Use Case Diagram

Flow:

1. **Login and Authentication**

The user with user credentials can log into the system.

A user is given access for authentication to verification services.

2. **Upload Currency Image**

Once authentication has been carried out, an authorized user can upload an image of a currency note to the website in jpeg or png format.

The system will verify against the acceptable types and sizes of images and simple quality requirements for those uploaded documents.

3. Input Validation and Integrity Checking

The uploaded images of the currency were tagged as corrupted or misformatted or showing unreadable characters by the system.

Invalidity of the input aborts processing by requiring a re-upload from the user.

4. Serial Number Extraction

The image of a currency serial number is input by image-processing techniques, commonly referred to as optical character recognition.

Also performed on these extracted characters are various preprocessing operations, inclusive of cleaning and character normalization.

5. Decision

Hotlist Check-Cross-checking of the extracted serial number with the locally maintained hotlist for identification of counterfeit currency finds the serial number on the hotlist.

The currency is immediately marked as counterfeit if it is found on the hotlist.

6. Blockchain Verification

In the absence of being blacklisted, the system will integrate the serial number to the verification through the blockchain process.

-Smart contracts will verify the numbers of serials through an immutable record in the ledger.

7. Decision of Verification

Whether the currency is genuine or is fake depends on the reply of verification from the blockchain.

8. Results Generation

The verified status would be a composite of results from local verification and from those the blockchain.

Logging and Audit Trail-Timestamp, serial number, verification results, and reference for transaction purposes are all included in the audit and traceability details.

Display Result to the User-Thus, through the system interface, the user receives the outcome of the verification (whether a genuine or counterfeit).

Explanation

The Activity Diagram illustrates the complete operational workflow of the proposed , beginning with user authentication and ending with the display of the final verification result. It highlights how the system processes currency images through a sequence of validation, serial extraction, blacklist checking, and blockchain-based authentication.

5.6 Sequence Diagram

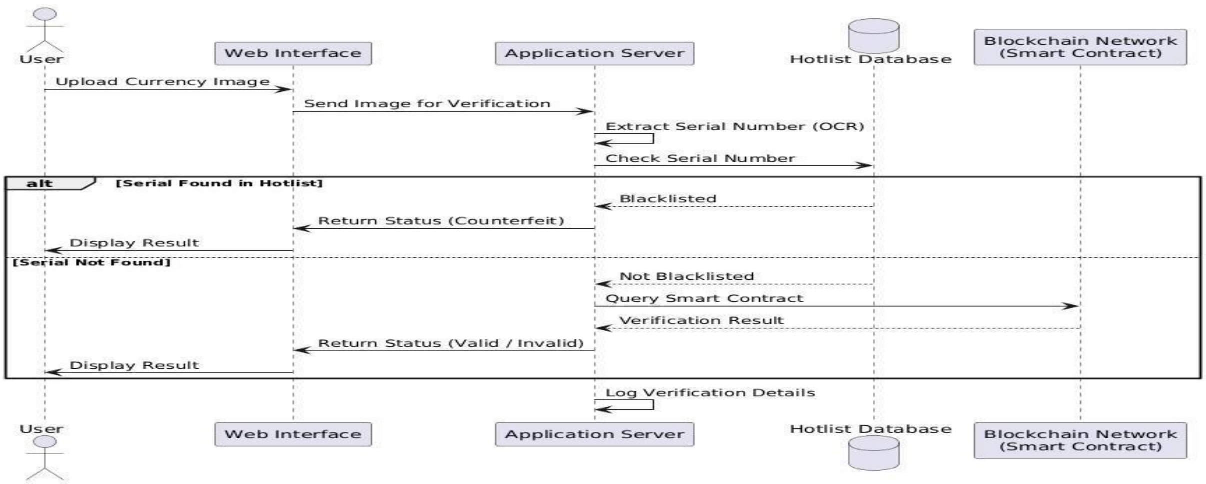

Fig 5.6.1 Sequence Diagram

Flow:

1. User → User Interface

- The user uploads a currency image (PNG/JPG) through the system interface to initiate the verification workflow.

2. User Interface → Application Server

- The interface forwards the uploaded currency image along with request metadata to the application server for processing.

3. Application Server → Input Validation Module

- The application server sends the image for validation to check file format, size, and basic integrity.

4. Input Validation Module → Application Server

- The validated image is returned to the application server for further processing.

5. Application Server → Serial Number Extraction Module

- The application server forwards the validated image to the OCR-based serial number extraction module.

6. Serial Number Extraction Module → Application Server

- The extracted and pre-processed currency serial number is returned to the application server.

7. Application Server → Hotlist Database

- The application server queries the hotlist database to check whether the extracted serial number is blacklisted.

8. Hotlist Database → Application Server

- The hotlist database returns the verification response indicating whether the serial number exists in the blacklist.

9. Application Server → Blockchain Network

- If the serial number is not blacklisted, the application server sends a verification request to the blockchain smart contract.

10. Blockchain Network → Application Server

- The blockchain network responds with the verification result (valid or invalid) based on cannot be modified ledger records.

11. Application Server → Result Processing Module

- The application server consolidates verification outcomes and prepares the final result.

12. Result Processing Module → User Interface

- The finalized verification status is sent to the user interface for display.

13. User Interface → User

- The user views the currency verification result and related details through the system dashboard.

Explanation

The Sequence Diagram further clarifies the temporal dependency between system components during the currency verification workflow. Each interaction is executed in a strictly ordered manner to ensure data integrity and controlled access to the blockchain network. The user initiates the process through the interface, but all sensitive operations—such as serial number extraction, blacklist validation, and blockchain querying—are performed exclusively by backend modules. This design prevents direct user interaction with the blockchain and minimizes security risks.

Methodology Explanation

1. Data Acquisition & Understanding

Users will be able to operate through the system interface and upload currency note images taken from the user's reference. The currency notes physically undergo a serial number as the main reference for verification. Since there are differences in the qualities of images-captured at different light, orientation, and backgrounds-the uniqueness of each of these factors comes into understanding to help ensure accuracy in extracting serial numbers

2. Data Preprocessing

Preprocessing is one of the stages where the raw currency images are potentially full of noise, distortion, shadows, and non-uniform formatting. This preprocessing stage has also provided cleaner, readable, and verifiable extracted serial numbers.

a. Image Validation

Uploaded images are validated first to determine that they are of acceptable format and size; those otherwise are rejected earlier in the process for unnecessary processing.

b. Noise Rejection

The techniques of image enhancement are incorporated to diminish the shadow noise from the image and read the clear characters in a more reliable manner for serial number extraction.

c. Serial Number Normalization

By removing non-essential characters, fixing the absence of spaces, and normalizing formats, extraction of a serial number can be analyzed and compared properly under such normalization.

d. Checks That Ensure Inputs Are Complete

These checks make the serial number derived from extraction perform checks on conformance to expected patterns in basic integrity. Thus, it avoids any entry of invalid data into the verification pipeline.

3. Logical Verification Architecture and Design

It incorporates a two-stage verification methodology with a view to keeping a balance between efficiency and trust.

a. Stage 1 Hotlist Verification

In the first stage, the extracted serial number is checked against a locally maintained hotlist containing known counterfeit or previously flagged currency identifiers. This stage enables rapid identification of known counterfeit notes and avoids unnecessary blockchain interactions.

b. Stage 2 Blockchain-Based Verification

If the serial number is not found in the hotlist, the system proceeds to blockchain verification. A smart contract deployed on the blockchain network is queried to validate the currency identifier against cannot be modified ledger records. This decentralized verification provides that authenticity data cannot be altered or manipulated by a centralized authority.

The separation of verification into two stages improves system efficiency while preserving the trust guarantees provided by blockchain framework.

4. Blockchain Interaction Mechanism

Blockchain framework is integrated into the system to address limitations of centralized verification systems. The blockchain serves as a decentralized and cannot be modified registry of currency identifiers.

5. Result Generation and Logging

After verification is completed, the system consolidates results from both local and blockchain checks.

The final verification status is generated.

Verification events, timestamps, and outcomes are logged for traceability and audit purposes. Logged data supports analysis of verification patterns and system behaviour.

Evaluation of Verification Outcomes

1. Verification Accuracy

Accuracy means what is in the records gets tested either to be true or false as far as a genuine currency is concerned. It is the correctness of the verification logic that is of concern as against any predictive sort of performance.

2. Response Time Observation

The system is also observed to quickly reject known counterfeit currency through hotlist checks. The latency of blockchain verification will also be studied to understand the feasibility in true real-time scenarios.

3. Consistency of Results

The same currency identifier, whenever verified, must return the same results for the workflow to be considered reliable.

System Logs and Observational Analysis

In this section, the verification behaviour will be analysed using system logs and observational data collected during testing.

1. Verification Event Logs

A log is maintained for serial number, verification stage, timestamp, and final result.

These logs serve now to further analyse the verification flow and detect possible anomalies that arise therein.

2. Blockchain Transaction Reference Analysis

The blockchain response is considered to ensure proper interaction with smart contracts and proper retrieval of verification results.

CHAPTER 6

MODEL DEVELOPMENT

When it comes to the present project, model development implies creation and execution of logical and functional models for currency verification through blockchain framework. Unlike traditional machine-learning projects, this system aims to build a secure verification model that integrates image processing, optical character recognition (OCR), cryptographic hashing, and blockchain-based record management.

6.1 Overview of the Verification Model

The currency verification model is designed as a multi-stage verification pipeline, where each stage performs a specific task and passes validated data to the next stage. The model does not rely on predictive learning but instead uses deterministic verification logic supported by blockchain immutability.

The core stages of the model are:

- Image acquisition and validation
- Serial number extraction using OCR
- Data normalization and hash generation
- Blockchain-based verification and logging
- Result classification and response generation

Each stage strengthens the trustworthiness of the final verification outcome.

6.2 Image Acquisition and Validation Model

The first stage of the model handles the acquisition of currency images from the user.

Image Validation Process

- Accepts only valid image formats such as PNG and JPG.
- Checks resolution and clarity to ensure OCR readability.
- Rejects corrupted or unsupported image files.

6.3 OCR-Based Serial Number Extraction Model

Optical Character Recognition (OCR) plays a critical role in identifying the serial number printed on the currency note.

OCR Workflow

- The uploaded image is converted into grayscale.
- Noise reduction and contrast enhancement are applied.
- OCR engine scans predefined regions where serial numbers are usually present.
- Extracted characters are cleaned using pattern validation.

Validation Rules

- Serial number length and format are checked.
- Invalid or incomplete serials are flagged.
- Only verified serial numbers are passed to the blockchain module.

This structured OCR approach minimizes false readings and extraction errors.

6.4 Hash Generation and Data Normalization

Once a valid serial number is extracted, it is normalized and converted into a cryptographic hash.

Hash Generation Logic

- Serial number, timestamp, system identifier are combined.
- SHA-256 hashing algorithm is used.
- The generated hash uniquely represents the verification event.

Hashing provides that:

- Sensitive data is not stored in plain text.
- Verification records cannot be altered.
- Each verification entry remains unique.

6.5 Smart Contract Logic (Backend Controlled)

Smart contracts provide instructions for the terms of approval and maintaining records.

Core Smart Contract Functions

- Add new currency verification record.
- Check existence of a serial number.
- Retrieve verification history.
- Generate cannot be modified transaction logs.

The smart contract provides:

- Automatic enforcement of verification rules.
- Resistant to alteration record handling.

No external wallet interaction is required for users.

6.6 Result Decision Model

After blockchain verification, the system generates a final decision.

Decision Output

- Verification status (Genuine / Suspicious)
- Transaction hash
- Timestamp of verification
- Blockchain block reference

This information is returned to:

- User interface for immediate feedback
- Admin dashboard for monitoring and auditing

6.7 Admin Monitoring and Log Model

The admin dashboard model provides real-time visibility into verification activity.

Admin Capabilities

- View all verification records.
- Track duplicate currency usage.
- Analyse verification trends.
- Inspect blockchain hashes and timestamps.

This model supports proactive fraud detection and system clear verification trail.

6.8 Security Considerations in Model Design

The layers below have been established in the system for security purposes:

- Simple Input Validation to prevent malicious uploads

- Cryptographic hashing to ensure the integrity of the data

- The administrative role-based access function

Thus, the above makes sure the integrity of the system is well fortified.

CHAPTER 7

MODEL EVALUATION AND TESTING

The most critical phases in evaluating an idea are model evaluation, testing, validation, and the proposition for application, to ensure the correctness, reliability, and effectiveness of the proposed currency verification system. This chapter further analyses the accuracy of the system in currency verification rates, the efficiency of duplicated or suspicious usages being detected, and reliability of records kept in the blockchain from being altered. Testing provides that the entire verification pipeline- from image upload to blockchain logging works properly under real world conditions.

7.1 Purpose of Evaluation

The evaluation process aims to confirm that:

- Currency serial numbers are extracted correctly using OCR.
- Blockchain verification logic works accurately for all cases.
- Duplicate and suspicious currency usage is detected reliably.
- Verification records are stored immutably and consistently.
- System responses remain stable under repeated and varied inputs.

This phase provides that the system is suitable for real-world deployment in financial verification environments.

7.2 Evaluation Parameters

The system is evaluated using the following parameters:

1. OCR Accuracy

- Measures correctness of extracted serial numbers.
- Evaluates performance under varying image quality and lighting.
- Confirms rejection of unreadable or invalid serial formats.

2. Blockchain Integrity

- Confirms immutability of stored records.
- Verifies that transaction hashes remain unchanged.
- Provides accurate timestamp and block mapping.

3. Response Time

- Measures time taken from image upload to verification output.
- Provides acceptable performance for real-time usage.

4. System Reliability

- Evaluates system behaviour during repeated usage.
- Confirms stability under multiple concurrent requests.

7.3 Test Scenarios

Multiple test scenarios were designed to validate different aspects of the system.

1. Valid Currency Verification

- Upload a clear currency image.
- Extract corrects serial number.
- Verify serial for the first time.

2. Duplicate Currency Detection

- Upload the same currency image multiple times.
- Serial number already exists on blockchain.
- Expected output: **Duplicate Currency Detected**.

3. Invalid Image Handling

- Upload blurred or unsupported image formats.
- OCR fails to extract serial number.
- Expected output: Verification Failed – Invalid Image.

Result Analysis

The tests proved that the system is reliable under any verifying conditions. Accuracy in OCR extraction is maintained for clear images while incorrect differentiates are rejected. In blockchain verification, duplicates and doubtful serial numbers were flagged without false positives.

The records of verification were found immutable through matching transaction hashes from several retrieval attempts. Response time is kept within acceptable limits for practical use.

Security and Reliability Testing

The security tests revealed the following:

The logs for verification cannot be altered or deleted.

Unauthorized users are blocked from admin features.

Sensitive information is secured using hashing and authorization.

Reliability testing demonstrated that the system is stable after prolonged actual verifications and repeated cycles over time.

Summary

Such a chapter examined the currency verification system as meeting all requirements both functional and security. Testing confirmed accurate serial extraction, reliable blockchain verification, effective duplicate detection, and reasonable system performance. Successful evaluation qualifies the system thus as a secure and trustworthy solution for preventing duplication and fraud in currency using the blockchain architecture.

CHAPTER 8

RESULT AND SNAPSHOT

The experimental evaluation of the proposed Currency Verification Using Blockchain system focuses on validating the feasibility, correctness, and operational reliability of the prototype. Since the system is designed as a research-oriented prototype, the evaluation emphasizes functional accuracy, response behaviour, and system stability rather than large-scale deployment performance. The experiments were conducted using controlled test inputs to simulate real-world currency verification scenarios.

The system successfully verified currency authenticity by combining image-based serial number extraction with blockchain-backed validation. Visual confirmation through the user interface, verification logs, and blockchain responses demonstrated that the proposed workflow operates consistently across multiple test cases.

8.1 Results

The results obtained from the prototype implementation demonstrate that the system performs the intended verification operations effectively. The evaluation primarily considers verification correctness, response flow, and interaction consistency among system components.

Key observations from the experimental results include:

- The system accurately extracts currency serial numbers from uploaded images under normal lighting and quality conditions.
- Known counterfeit serial numbers stored in the hotlist are identified and rejected at an early stage.
- Blockchain verification is triggered only when required, validating the efficiency of the two-stage verification design.
- Verification outcomes remain consistent across repeated test executions.

These results collectively validate the feasibility of using blockchain for secure currency verification in a prototype environment.

8.2 Comparison with Existing Methods

Criteria	Existing Centralized Methods	Proposed Blockchain-Based System
Verification Storage	Centralized database maintained by a single authority	Decentralized immutable ledger that cannot be altered
Data Integrity	Susceptible to modification or deletion	Tamper-resistant records ensured through cryptographic hashing
Trust Dependency	Relies on a single trusted authority	Distributed trust across multiple blockchain nodes
Auditability	Limited transparency and audit capability	Fully transparent and traceable transaction history
Verification Speed	Moderate due to centralized processing	Optimized using hotlist filtering combined with blockchain verification
Scalability	Limited by central server capacity	Conceptually scalable due to decentralized architecture
Security	Vulnerable to insider threats and single-point failures	Strong cryptographic assurance and resistance to attacks

8.3 Snapshots

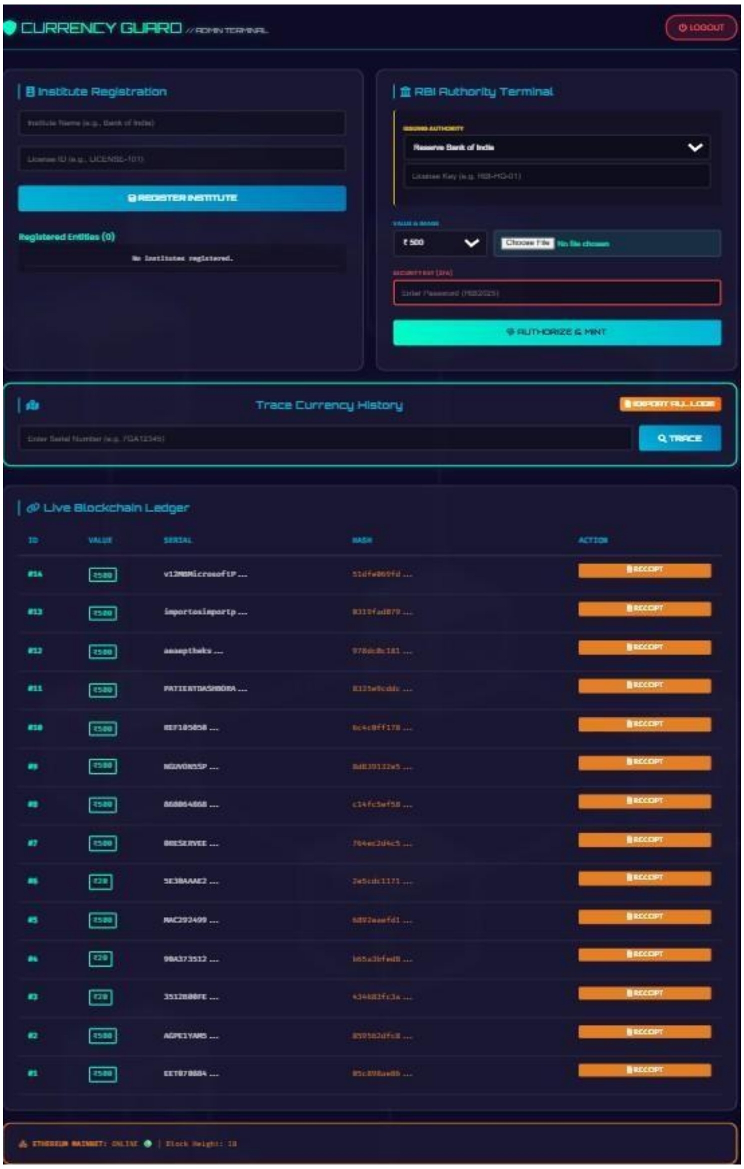

Fig 8.3.1 Admin Dashboard

The Admin Dashboard Interface provides administrative access to monitor and manage the currency verification system. It allows the administrator to review verification logs, observe blacklisted serial numbers, and track system activity generated during verification workflowes. This dashboard supports oversight and audit functions by presenting verification records and system status in an organized manner. Although implemented at a prototype level, the admin interface demonstrates how centralized control and monitoring can complement decentralized blockchain-based verification.

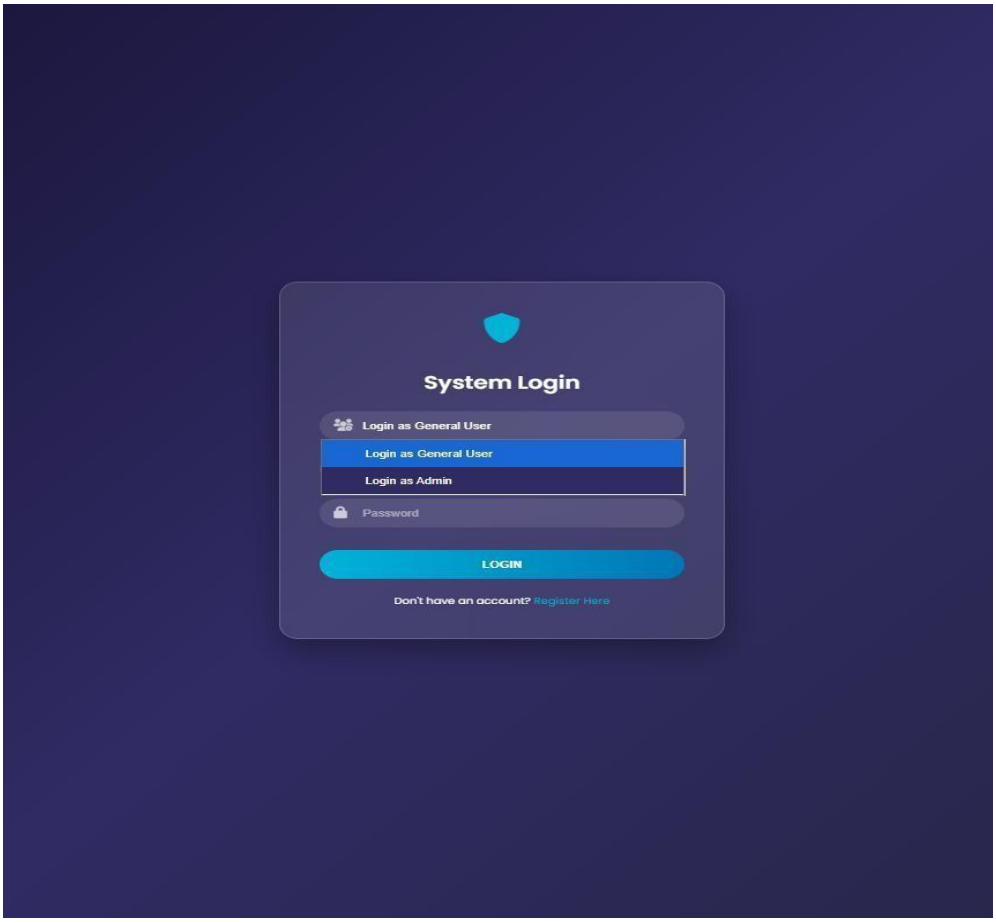

Fig 8.3.2 Login Page

The system begins with a secure login interface where authorized users enter credentials to access verification services. This step provides controlled access and prevents unauthorized usage of the system.

8.3.3 Currency Image Upload Section

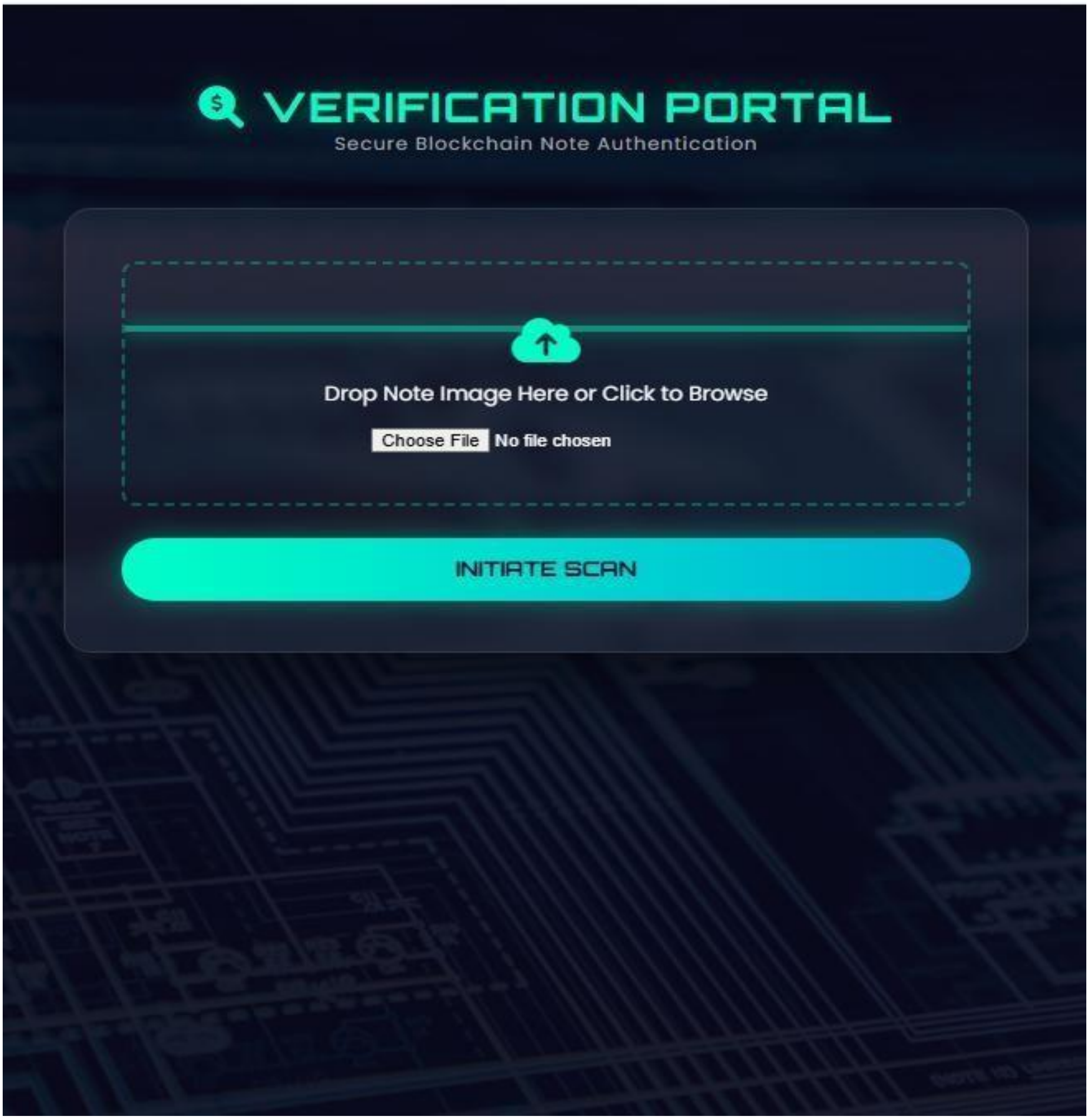

Fig 8.3.3 **Currency Image Upload Section**

The dashboard acts as the entry point for all verification activities and integrates the OCR module, verification logic, and blockchain interaction. This interface demonstrates the practical usability of the prototype and enables seamless interaction between the user and the underlying verification workflow.

8.3.4 Verification Completion Display

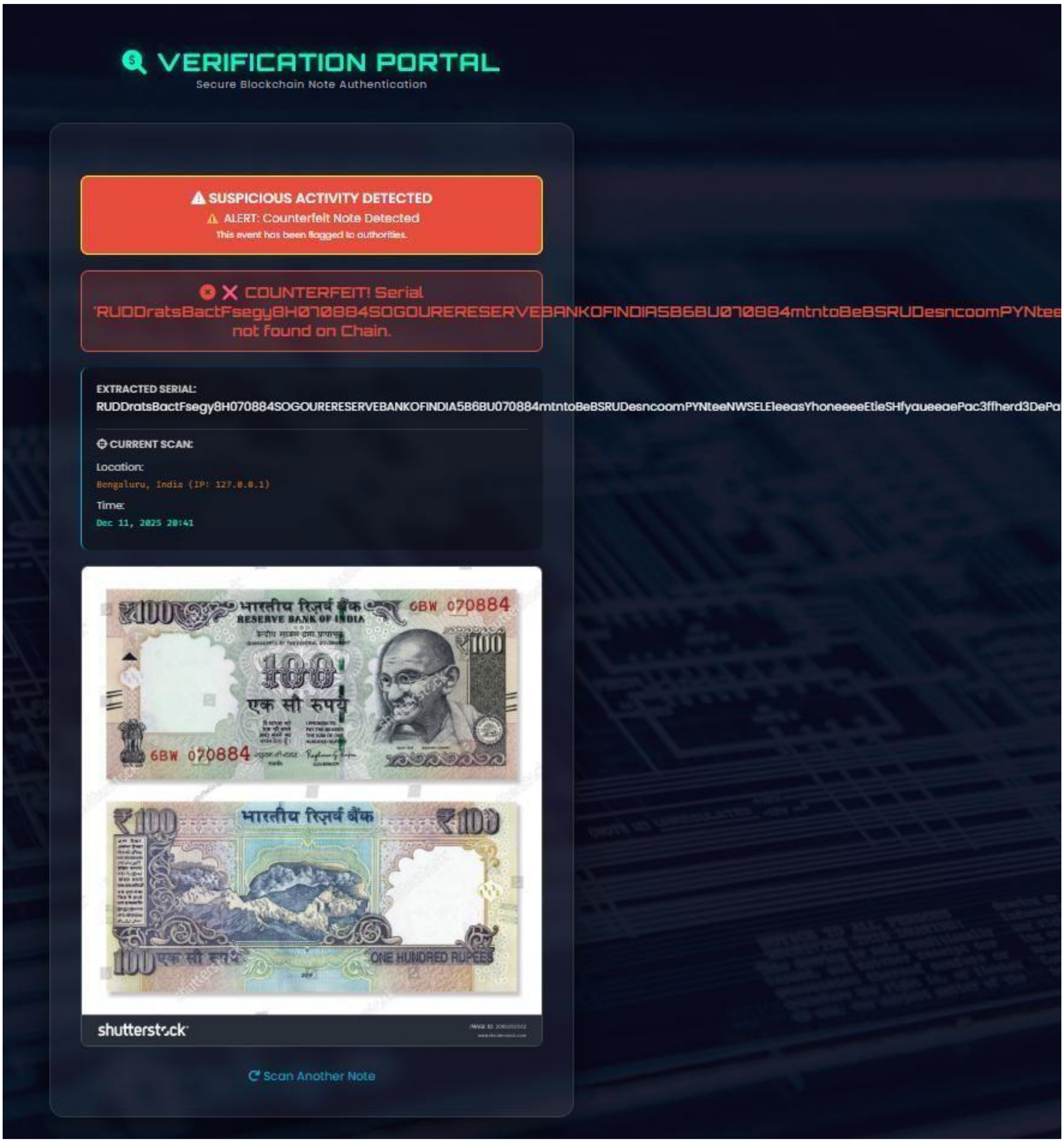

Fig 8.3.4 Counterfeit Currency

Fig 8.3.5 Genuine Currency

8.3.6 Blockchain Verification Ledger View

Fig 8.3.6 Blockchain Verification Ledger View

8.4 Observation and Analysis

The experimental observations indicate that The developed system operates reliably under prototype conditions. The two-stage verification approach significantly improves efficiency by reducing unnecessary blockchain queries. Hotlist-based filtering provides rapid rejection of known counterfeit currency, while blockchain verification provides high-assurance validation for unknown cases.

Analysis

1. **Verification Accuracy**

   The system correctly classifies currency based on available records, with no contradictory results observed during testing.

2. **Efficiency of Two-Stage Verification**

   Early rejection through hotlist verification reduces processing time and blockchain dependency.

3. **Blockchain Effectiveness**

   Blockchain provides immutability and trust, addressing limitations of centralized verification systems.

4. **System Stability**

   Repeated tests confirm consistent behaviour without unexpected failures.

CONCLUSION

The main aim of this project was to study and demonstrate whether blockchain framework can be practically used as a reliable method for currency verification. Existing currency authentication methods mainly depend on physical security features or centralized databases. Although these methods are widely used, they have several drawbacks such as vulnerability to tampering, limited clear verification trail, and complete dependence on a single authority. To overcome these issues, this project explored an alternative approach by combining image-based serial number extraction with blockchain-supported verification.

Very fundamentally, architecture has a criteria clearly implemented verification workflow which links conventional image processing to distributed ledger technology. From the two-stage validation process, currency serial numbers are extracted from the uploaded images. The first hotlist check is a rapid way of capturing known counterfeit notes while the second one is

confirmation through the blockchain, which offers it up as a general case basis for being authentic. Thus speed in conducting verification processes can be achieved without sacrificing the reliability and integrity of the data.

Most importantly, this research asserts and concentrates on creating the basis for a blockchain framework. It will thus make it a trust-enforcing layer as opposed to enforcing the entire system with the blockchain. This design choice goes towards mitigating some of the running problems faced in a centralised system, such as data-tapping and unaccountability. Negative test results would imply that the verification records stored in the block are intact, not tampered with, and, thus, confidence in the verification workflow may be raised.

This project, therefore, handles system convenience during designs in modules. Major operations like image preprocessing, hotlist checking, serial number extraction, blockchain interaction, and result generation will be considered as different modules. Such a strategy promotes clarity and maintainability for future enhancements without entangling them into the whole system. Their results were similar over repeated successive testing, thus proving the stability and correctness of the system architecture.

FUTURE ENHANCEMENTS

Through Currency Verification Using Blockchain, the possibility for currency authentication using blockchain within a prototype environment shows some room for improvement. Evaluating correctness and feasibility seems to be the main purpose of the current implementation under consideration. In the future, it may further improve efficiency, scale, and real-life usability, but without making any bold claims of readiness for deployment. One possibility is improving the extraction of serial numbers. Presently, fairly rudimentary optical character recognition (OCR) systems operate well under real-world conditions; however, this ability is greatly influenced by the low quality of images taken for detection, unanticipated lighting conditions, or physical degradation of currency notes. Future work in this context may begin to standardize the use of deep learning techniques-whether for audiovisual or text recognition models-in line with the image processing paradigms stated.

The other major area to be improved is the image-preprocessing techniques. Advanced preprocessing methods such as illumination normalization, perspective correction, and background noise reduction may be incorporated in the next phase of the project. This would ensure the system would be robust against pictures taken in uncontrolled surroundings while minimizing errors during extraction of serial numbers, thereby enhancing overall verification reliability. Future works could also propose design alternatives to the smart contract itself for optimization. In such a scenario, it is anticipated that a better smart contract would create less computational overhead while providing a quicker response. Further, some versioning schemes can be regarded as an appropriate way of updating the verification logic in a manner that does not disturb already stored records, thus ensuring the blockchain layer is maintained through time.

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